Lightweight Classification Models for Automated 5-stage Sleep Scoring from EEG Data

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1 Abstract

Automated sleep stage classification using electroencephalography (EEG) remains a critical bottleneck in computational neuroscience due to substantial inter-subject biophysical variability. While high performance is routinely reported in subject-dependent validation paradigms, these architectures frequently fail to generalize across alien subjects. In this study, we present a rigorous, high-powered benchmarking framework utilizing an expanded cohort of 30 healthy subjects extracted from the open-access Bitbrain Sleep (BOAS) dataset on OpenNeuro. We systematically evaluate six distinct machine learning and deep architectures—including eXtreme Gradient Boosting (XGBoost), Random Forest, and Feedforward Neural Networks (FNN)—across an engineered 96-feature EEG matrix. To map the boundary conditions of personalization versus universality, models were rigorously cross-validated using a three-tier paradigm: true Within-Subject 10-fold Cross-Validation, Pooled Subject-Dependent K-Fold, and Subject-Independent Leave-One-Subject-Out (LOSO) Cross-Validation. Evaluation criteria prioritize class-balanced metrics, utilizing Cohen's Kappa (κ) and Macro F_1-score to dissect specific architectural failures on the highly non-stationary and minority transitional N1 sleep stage. Our results demonstrate that tree-ensemble architectures yield superior cross-subject performance and performance's stability, demonstrating that precise spectral feature engineering significantly mitigates the geometric and biophysical drift observed across distinct subjects.

2 Introduction

Classifying sleep stages is crucial for diagnosing sleep disorders like insomnia, OSA (obstructive sleep apnea), Hypersomnia and parasomnia. In spite of this, scoring is still mostly done manually by Polysomnographic technologist (PSG technologists) following the criteria issued by the AASM (American Academy of Sleep Medicine) [1]. This is of course very expensive and time consuming, and introduces a substantial degree of subjectivity, which needs to be mitigated by employing more than one technologist to score the recordings, making the process even more expensive. Another aspect that is worth noting is the complexity of the problem: sleep patterns and characteristics have a great degree of inter-individual variability, it is therefore crucial to keep into account individual differences, and this is the strongest factor limiting automated scoring systems development and adoption. Aside from this variability, there is also another aspect involved: the PSG machine itself. Depending on the machine, it can have a variable number of downtimes, produce artifacts, be imprecise in the recording, and introduce noise. Given all these factors and circumstances,

even if the various sleep stages are well defined and separated on paper, in practice there is a good degree of overlap, and it is often difficult to be sure on which stage each particular epoch belongs to.

When building an automatic scoring system, there are 3 main characteristics that are desirable: high accuracy, low complexity, and efficiency. Low complexity is desired for explainability reasons: being a medical problem, it is important for the model to be transparent and explainable. Efficiency is mainly desired when building models that will be used to implement real-time sleep monitoring systems, which allow patients to monitor their sleep quality directly from their homes. To this date, there is still no model that fully satisfies all these 3 requirements.

Our aim for this project was to find the best low complexity model, one that could achieve competitive accuracy ($> 70\%$) on generalization tasks using only EEG recordings, while still being fast, efficient, and therefore employable in real-time monitoring systems. We constructed models of increasing complexity, the simplest being KNN, and the most complex being a simple FeedForward Neural Network. We then evaluated their performance on the data to find the best one in terms of trade-off between complexity and accuracy. Given that based on the application models will have different requirements, we applied 3 different cross-validation techniques when testing the models: Leave-One-Subject-Out (LOSO) cross-validation, pooled 10-fold cross-validation and within-subject 10-fold cross-validation. The first two test generalization abilities, while the last one tests personalization abilities.

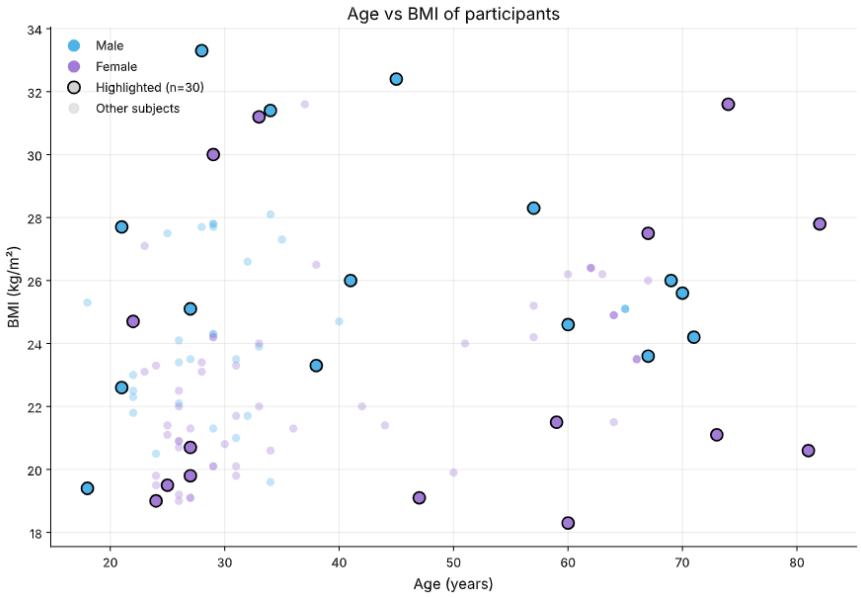


Figure 1: Selected participants descriptions: shows a scatterplot representing each participant as a dot with 3 dimensions, age on x-axis, bmi on y-axis and gender encoded as color. Highlighted subjects are the 30 we selected.

3 Background

3.1 Sleep Theory

Sleep is traditionally divided into two main types: REM sleep and Non-REM sleep, REM sleep is characterized by rapid movements of the eyes, from which the stage gets its name (REM: rapid eye movements), while Non-REM sleep is further divided into 3 more substages: Non-REM 1, Non-REM 2 and Non-REM 3. Other divisions have been proposed in the past but this is the most widely accepted.
Non-REM 1 (N1) is the lightest stage of sleep and typically marks the transition from wakefulness to sleep. EEG patterns during this stage show a reduction in alpha activity (8–12 Hz), while low-amplitude, mixed-frequency theta waves (4–7 Hz) usually show a gradual increment. Eye movements are slow and rolling, and muscle activity begins to decrease. Heart rate and respiration are slightly slower than in wake stage.
Non-REM 2 (N2) is a deeper stage. In this stage, conscious awareness of the environment fades completely. It is characterized by the presence of sleep spindles (brief bursts of 12–16 Hz activity) and K-complexes (sharp, high-amplitude waveforms) in the EEG. These phenomena are believed to play roles in memory consolidation and sleep maintenance. Eye movements stop, muscle tone and autonomic functions like heart rate and breathing continue to decrease.
Non-REM 3 (N3), also known as slow-wave sleep or deep sleep, is the most restorative stage. Activity is dominated by delta waves (0.5–2 Hz), which are high in amplitude and low in frequency. This stage is associated with minimal responsiveness to external stimuli, further reductions in heart rate, blood pressure, and breathing, as well as almost no eye movement and very low muscle activity.
Finally, REM sleep shows an EEG pattern similar to wakefulness or light sleep, in spite of the person being in a deep sleep state. This activity is accompanied by rapid eye movements, vivid dreaming, and a near-total loss of muscle tone (atonia), which supposedly is in place to prevent individuals from acting out their dreams. Heart rate and breathing become more irregular and may increase compared to Non-REM stages[1].
Sleep cycles through these stages during the night. Each cycle lasts about 90 minutes, but the dominance of each stage during these 90 minutes varies greatly between cycles, with N3 sleep being more dominant in early cycles and progressively reducing its duration as the night progresses in favor of lighter non-rem stages and REM sleep.[2]

3.2 Summary of Literature

In the last 10 to 15 years, a wide range of machine learning techniques have been employed to approach the problem of automatically classifying sleep stage. Earlier attempts implemented mostly traditional models like SVMs, K-Nearest Neighbors, Random Forests or LDA, trained on features extracted from EEG, EOG and EMG signals. In general, the final feature vector included a mix of statistical, spectral or entropy-based features. These models showed promising results in binary or three-class tasks, in which they achieved very high accuracies, sometimes exceeding 90%, but still struggled with the full 5-class classification problem, where accuracy tends to drop below 80%, even in studies where multimodal datasets have been constructed [6]. With the advent of Neural Networks, deep learning approaches have taken over, with CNNs and RNNs (particularly LSTMs and GRUs) showing much better performances (at the cost of interpretability, computational cost and increased complexity) [4]. CNNs can learn spatial patterns directly from the raw signal, while RNNs can capture the temporal dependencies between epochs, hence the better performances. Some of the best-performing models combine both, this is the case for example of DeepSleepNet

[7] or SeqSleepNet [5]. Both report accuracies above 85% and Cohen's kappa scores close to 0.8. These models, however, are computationally expensive and often lack interpretability. The most recent approaches are exploring attention mechanisms, temporal smoothing via CRFs, and subject adaptation techniques- The results for these models up to now seem solid, but unfortunately, their high complexity makes them unsuitable for deployment in real-time or explainable systems. As discussed in the introduction, while deep learning models largely dominate in terms of raw accuracy, we are still lacking a model that could represent a good trade-off between accuracy, speed and interpretability.

4 Data and EDA

4.1 Data Description

The Data we worked on comes from the BitBrain Open Access Sleep (BOAS) dataset [3], openly available on OpenNeuro. The full dataset includes 128 full nights recordings from 108 subjects, recordings were done using 2 different machineries, PSG and an EEG headband, PSG recordings include 6 EEG channels (frontal left, frontal right, central left, central right, occipital left and occipital right), 1 EOG channel (electrooculography), 1 EMG channel (electromyography), 1 ECG channel (electrocardiography), 1 PPT channel (Photopletismography), plus recording of breathing activity and air flow. All the nights were divided in 30-seconds long epochs (around 1000 epochs per night) and scored both manually and autonomously (ML model, indicated as "scoring-AI"). For all subjects, basic demographic data were available (age, sex, BMI). Given the great volume of data contained in the dataset and our limited hardware capabilities, we decided to work on a subset of 30 subjects, carefully selected to maintain the original dataset's distribution and variability, out of those whose data contained all the required channels (as not all subjects had data for all channels). Therefore, we selected 15 men and 15 women across a wide range of ages and BMIs.

4.2 EDA

We performed Exploratory Data Analysis on our Dataset to get insights and to understand basic patterns contained in the Data. The following paragraphs summarize relevant information that emerged from our analysis.

4.2.1 Inspection of stage distributions:

The EDA revealed a severe class imbalance inherent to natural sleep architecture, as shown in Figure 2 (top left). The N2 stage constitutes almost 60% of the data, while the transitional N1 stage is significantly underrepresented. We can also notice that some subjects, specifically subject 3, 6, 10, 12, 13, 14, 16, 18, 19, 21, 24, 26 and subject 30 of our subset, never entered deep sleep (N3), which can happen, for example, when the person is not sleeping well. Epochs labeled as disconnections were removed.

4.2.2 Stages transition matrices:

Transition matrices gave us a clear overview of how on average sleep might progress through the stages. For the purpose of our work, the most relevant information we extracted from these matrices is that for each epoch belonging to a certain stage, there was a high probability that the next epoch would belong to the same class as well. However, this holds for all classes except N1, which has a

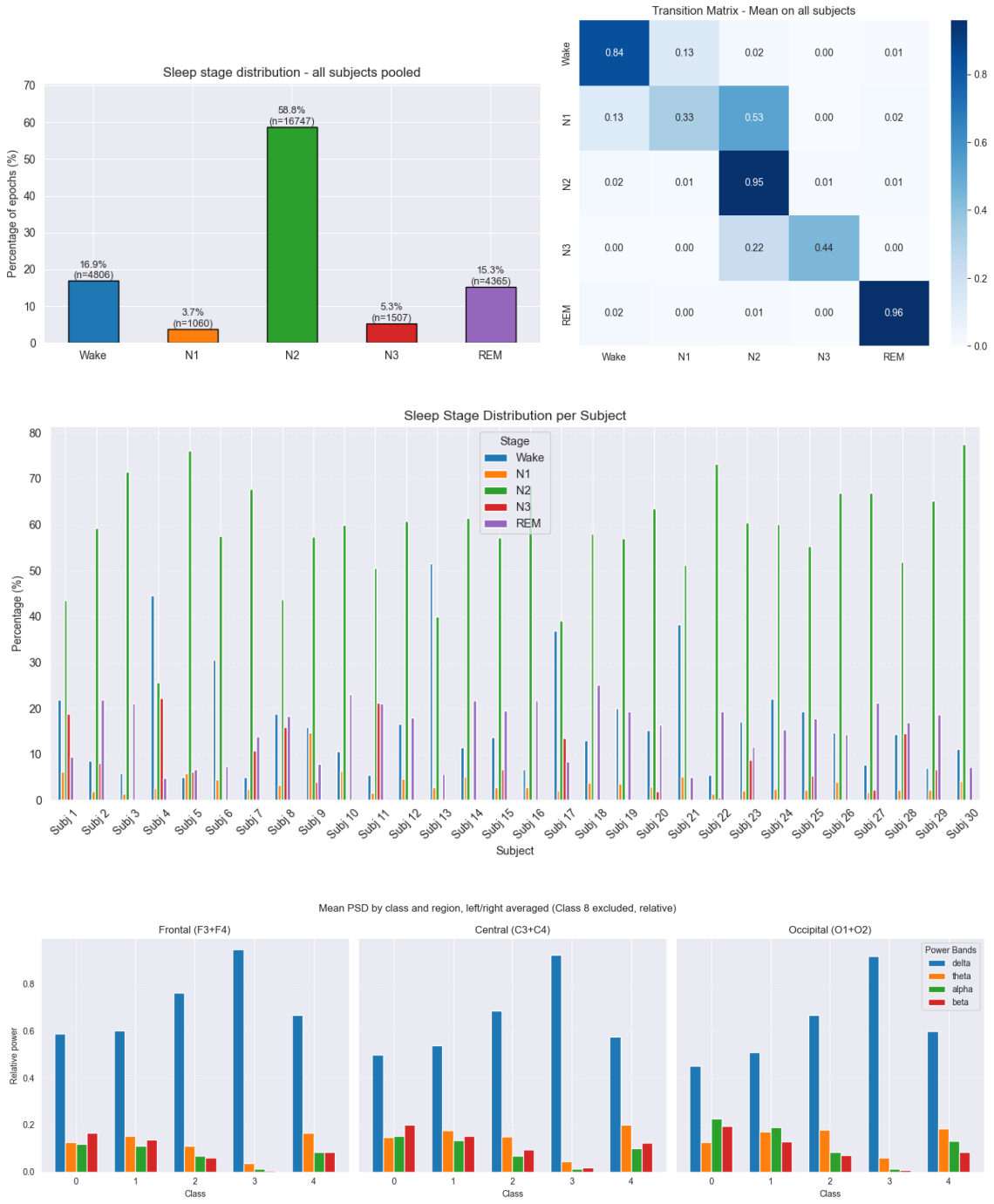


Figure 2: Top left and center: Stage distributions. Top-left shows the total sleep stage percentage over all subjects, center shows the percentage of each stage during sleep for each subject. Top right: Stage Transition Matrix. Mean transition matrix for all subjects. Bottom: psd analysis. This graph shows for all classes the mean power for each band aggregated by area and averaged over all subjects.

higher probability that the next epoch will belong to N2 (>50%), and a quite relevant probability (about 13%) that the next epoch would belong to wake, equal to the probability that the epoch following an epoch belonging to awake will belong to N1. Another interesting fact worth mentioning is that from N3 we have almost a quarter of chance that the next epoch will belong to N2. The only classes for which we can be almost sure that the next epoch will be of the same class are N2 and REM, for which there's a 95% transition chance.

4.2.3 PSD (Power Spectral Density) Analysis:

The psd revealed some other interesting insights that really help interpreting some results later seen from the classification models, also partially explaining the reason behind the most common misclassifications. As we can see, the awake class (class 0) is the one that has the highest mean powers for bands beta and alpha, which complies with the literature because these bands are, as previously discussed in the background section, high frequency or fast waves, and are mainly produced when the brain is active. Given the setting of the experiment, the subjects were most likely relaxed, which explains why alpha and beta bands are not dominant for this class, even though they're more present than for other classes. We can also notice how N3 (class 3) is the class that has the highest mean powers for delta and theta bands. Given that these bands are slow wave bands and N3 is deep sleep stage, this observation also finds support in the theoretical background. Aside from this, it is also interesting to notice how, depending on the channel, N1 looks similar to either wake or N2, which partially explains the difficulties some of our models encountered when classifying N1, as we will later see.

5 Methods and Development

5.1 Preprocessing and Feature extraction:

We started our preprocessing pipeline by loading the EDF files containing polysomnographic EEG recordings and corresponding scoring files that provided annotated sleep stages. We selected only the six EEG channels (F3, F4, C3, C4, O1, O2) and applied a band-pass FIR filter between 0.5 and 40 Hz to remove low-frequency drifts and high-frequency artifacts. From each scoring file, we extracted sleep epochs by identifying the start and end sample indices of each annotated segment and retrieving the corresponding data slices from the filtered signal. For each channel, we extracted statistical descriptors such as mean, standard deviation, variance, skewness, kurtosis, and root mean square (RMS), as well as spectral power in four frequency bands (delta, theta, alpha, beta) using Welch's method. We also computed all pairwise ratios between these bands. Given 6 EEG channels, this resulted in a total of $6\times6 = 36$ statistical features, $6\times4 = 24$ spectral power features, and $6\times6 = 36$ band ratio features, totaling 96 features per epoch. We trained and tested all of our models on this version of the dataset.

5.2 Evaluation Approach:

As mentioned previously, classification models of sleep stages can be used for a wide variety of applications: from real-time sleep monitoring and automated sleep scoring in sleep clinics, to individual-level applications such as personalization of therapeutics. These different applications have different requirements and may prioritize performance on different metrics other than sheer accuracy. We therefore decided to test our models on 3 different tasks and cross-validations. First, we tested on LOSO cross-validation because it allowed us to evaluate each model's generalization capabilities.

Second, we tested on pooled 10-fold cross-validation, because it allowed us to evaluate each model's generalization capabilities in a group setting. This can be seen as a mid-level generalization and might be useful, for example, to evaluate whether a model will be applicable to automate new data scoring of groups of patients from which data had already been collected and scored on previous occasions. Lastly, we tested on within-subject 10-fold cross-validation, because this allowed us to evaluate each model's personalization capabilities, as this is crucial for building accurate individualized monitoring systems.

5.3 Development:

5.3.1 Common setup:

All models were trained on the same $p = 96$ features extracted from each epoch, after discarding the epochs of class 8, which leaves the five stages Wake, N1, N2, N3 and REM. Features were standardized with a z-score whose mean and standard deviation were computed on the training portion of each split only and then applied to the corresponding test data, so that no information from the held-out subject or fold entered the scaling. No class rebalancing or class weighting was applied; class imbalance is instead accounted for by reporting Cohen's κ and macro-F1 along with accuracy. When needed, model settings were defined during an initial development phase on a stratified 80/20 hold-out split and then kept fixed across the three evaluation schemes (LOSO, pooled 10-fold and within-subject 10-fold), without any scheme-specific tuning.

5.3.2 XGBoost:

Gradient boosted trees were trained with the `xgboost` package using the multiclass objective `multi:softprob` and 500 boosting rounds, with all other hyper-parameters at the package defaults (e.g. learning rate 0.3 and maximum depth 6). The predicted class is the one with the highest predicted probability. No early stopping was used.

5.3.3 FNN:

The FNN was implemented in `torch` for R and consists of two fully connected hidden layers with 130 and 16 units and ReLU activations, followed by a linear output layer with five units (the softmax is included in the cross-entropy loss). It was optimized with Adam (learning rate 10^{-3}) using full-batch updates for at most 2,500 epochs. To limit overfitting, we used early stopping: 20% of the training data, stratified by class, were held out as an internal validation set, validation accuracy was monitored at every epoch, training was stopped after 300 epochs without improvement, and the weights of the best epoch were restored. The test data was never used for early stopping.

5.3.4 Random Forest:

The random forest was fitted with the `ranger` package using its default settings: 500 trees, $\lfloor \sqrt{p} \rfloor = 9$ candidate features at each split, and fully grown trees. No hyperparameter tuning was performed.

5.3.5 KNN:

KNN was implemented with `class::knn` on standardized features (Euclidean distance) with $k = 7$. The value of k was explored by leave-one-out cross-validation (`knn.cv`) on the training set, comparing odd values from 1 to 11. Since KNN has no training phase, the training time reported for this model corresponds to the time needed to predict the test epochs.

5.3.6 Linear and Radial SVM:

Support vector machines were fitted with the `e1071` package (C-classification) using either a linear or a radial basis function kernel. Since features were already standardized, the package's internal scaling was disabled. Hyper-parameters were left at their default values ($C = 1$ and, for the radial kernel, $\gamma = 1/p$), without tuning. To reduce computational cost, the fits for different subjects and folds were run in parallel on separate processes.

6 Results

Table 1: Leave-one-subject-out cross-validation (30 folds). Values are mean ± standard deviation across folds; models sorted by accuracy; best value per metric in bold. Best performing models highlighted in grey.

Model	Accuracy	Cohen's κ	Macro-F1	Training time (s)
XGBoost	**0.767** ± 0.082	**0.576** ± 0.144	0.596 ± 0.102	177
Random Forest	0.748 ± 0.100	0.539 ± 0.153	**0.647** ± 0.112	70.0
FNN	0.724 ± 0.140	0.528 ± 0.173	0.583 ± 0.141	44.0
SVM (RBF)	0.719 ± 0.135	0.488 ± 0.184	0.621 ± 0.117	994
SVM (linear)	0.718 ± 0.152	0.488 ± 0.203	0.615 ± 0.148	1315
KNN	0.665 ± 0.130	0.409 ± 0.163	0.498 ± 0.128	**3.3**

Table 2: Pooled 10-fold cross-validation (epochs from all subjects shuffled together). Values are mean ± standard deviation across folds; models sorted by accuracy; best value per metric in bold. Best performing models highlighted in grey.

Model	Accuracy	Cohen's κ	Macro-F1	Training time (s)
XGBoost	**0.899** ± 0.003	**0.828** ± 0.004	**0.784** ± 0.006	160
Random Forest	0.887 ± 0.003	0.806 ± 0.006	0.753 ± 0.010	62.0
FNN	0.845 ± 0.007	0.737 ± 0.011	0.693 ± 0.019	44.5
KNN	0.800 ± 0.004	0.652 ± 0.007	0.650 ± 0.007	**8.8**
SVM (linear)	0.786 ± 0.007	0.618 ± 0.013	0.743 ± 0.014	1167
SVM (RBF)	0.784 ± 0.009	0.612 ± 0.017	0.738 ± 0.016	825

Table 3: Within-subject cross-validation (one model per subject, 30 subjects). Values are mean ± standard deviation across subjects; training time is the mean per-subject time; models sorted by accuracy; best value per metric in bold. Best performing models highlighted in grey.

Model	Accuracy	Cohen's κ	Macro-F1	Training time (s)
XGBoost	**0.904** ± 0.036	**0.825** ± 0.051	0.774 ± 0.046	3.22
Random Forest	0.903 ± 0.035	0.823 ± 0.051	0.788 ± 0.061	0.38
SVM (linear)	0.884 ± 0.039	0.790 ± 0.061	0.734 ± 0.055	0.14
FNN	0.881 ± 0.038	0.784 ± 0.058	0.746 ± 0.074	6.22
SVM (RBF)	0.872 ± 0.046	0.761 ± 0.085	**0.807** ± 0.083	0.32
KNN	0.869 ± 0.046	0.759 ± 0.068	0.749 ± 0.072	**0.01**

6.1 Evaluation Metrics:

Because of the nature of the problem, we decided to evaluate the models based on the following metrics: Accuracy, Macro F1 and Kappa Value. Accuracy has been chosen because it enables us to evaluate the rate of correct classifications over total. It is the most important metric because it gives us the most immediate, interpretable and clear overview of the performance of a model before proceeding with further evaluation. Macro F1 is the average of the F1 scores for all classes for a given model. This metric has been chosen because it allows us to easily evaluate both precision and recall for the entire model after taking a look at individual F1 scores for each class, almost acting as a summary. Lastly, the Kappa Value (or Cohen's Kappa), which is a metric that measures the agreement between the model's predictions and the true labels, while accounting for the agreement that could happen by chance. Unlike accuracy, it considers the imbalanced distribution of classes. We chose it because it provides a more robust and reliable evaluation, especially when dealing with imbalanced datasets, where accuracy alone can be misleading.

6.2 XGBoost:

XGBoost emerged as one of the top-performing models in our analysis, achieving an average accuracy of 76.7% in the LOSO cross-validation, an accuracy of 89.9% in the pooled 10-fold cross-validation, and an accuracy of 90.4% in the within-subject 10-fold cross-validation. However, it showed quite some variability in its performance under LOSO cross-validation, as demonstrated by its accuracy's standard deviation of 0.08. Nonetheless, it is the most consistent out of the models we tested in both LOSO and pooled 10-fold cross-validation. It also showed the highest κ-values and achieved decent Macro-F1, although Random Forest surpassed it on this metric in 2 out of 3 cross-validation modes. Its success can be attributed to its powerful gradient boosting architecture, which effectively handles complex interactions within the feature set, and its built-in regularization, which certainly helped generalizing across different subjects. While it demonstrated strong performance on stable sleep stages like N2 and Wake, like all other models it struggled with the accurate classification of the transitional N1 stage.

6.3 FNN:

FNN (Feedforward Neural Network) is only our third best performing model out of the block. When tested using pooled 10-fold cross-validation, it achieved an average accuracy of around 84.5%, macro F1 scores around 70%, and κ-value around 74%. Under LOSO cross-validation, it achieved a good accuracy of 72.4% on average, although it also showed the second highest variability in performance, as demonstrated by its standard deviation of 0.14. Under within-subject cross-validation testing, it was surpassed by the linear SVM. It achieved an accuracy of 88.1% against the 88.4% achieved by the linear SVM. However, it managed to maintain a higher Macro-F1 than the linear SVM. As for other models, its main Achilles' heel is class N1, which is almost completely ignored. We can clearly see this from the extremely low accuracies for that class, with some folds having an accuracy of 0%, as well as the terrible F1 scores, which in most cases falls below 30%.

6.3.1 Random Forest:

Random Forest proved itself competitive with XGBoost, achieving very close accuracies with better F1-scores in 2 out of 3 of the cross-validation we tested, while needing much less training time. Under LOSO cross-validation testing conditions, it achieved an accuracy of 74.8% with standard deviation of 0.1, similar to what XGBoost achieved, though with a slightly lower κ-value (54%)

and a slightly higher Macro-F1 (64.7%), which shows that this model dealt with class imbalance slightly better than XGBoost did. On the Pooled 10-fold cross-validation it achieved an accuracy of 88.7%, just below that of XGBoost, while on the within-subject 10-fold cross-validation, it achieved almost the exact same accuracy as XGBoost, but with slightly lower standard deviation and higher Macro-F1. Like XGBoost, this model also struggled to correctly classify N1 epochs, though less than other models did.

6.3.2 KNN:

Interestingly, KNN achieved unexpectedly good accuracies and Macro-F1 scores. While it turned out to be the worst performing model under LOSO testing, it achieved an accuracy of at least 80% under both 10-fold cross-validation conditions, with on average macro-F1 score higher than that of the FNN under within-subject cross-validation (74.9% against FNN's 74.6%, with lower standard deviation as well). This model also has the big advantage of not needing any training. Its parameter k must be tuned before-hand, but once that is done, predictions are available right away.

6.3.3 Linear and Radial SVM:

Lastly, we built a radial SVM and a linear SVM. The linear SVM achieved higher accuracies than the radial SVM on both the pooled and within-subject 10-fold cross-validations (accuracy of 78.6% vs 78.4% and 88.4% vs 87.2%), while the radial SVM slightly outperformed the linear SVM under LOSO conditions, and achieved a higher macro-F1 under both LOSO and within-subject cross-validation, where it actually achieved the highest Macro-F1 score out of all models. However, under LOSO and pooled 10-fold cross-validation, their training time was on the order of 5 minutes (15 or plus minutes when parallelized), which might limit applicability in some contexts.

7 Discussion and Conclusions

The results obtained throughout this project reflect both the potential and limitations of using low-complexity models for sleep stage classification. All our models performed well above the chance level (20%), which suggests that even basic models like KNN can pick up most stage-specific characteristics, particularly for more distinct stages like Wake or N3. XGBoost and Random Forest performed significantly better than all other models under all 3 testing conditions, with XGBoost surpassing Random Forest under Pooled 10-fold cross-validation and Random Forest winning the trade-off under within-subject 10-fold cross-validation. Under LOSO conditions, they achieved basically the same performance, with XGBoost achieving an only slightly higher accuracy and κ-value, especially when considering that the difference between the two models' performances is smaller than their standard deviation, while Random Forest achieved the highest Macro-F1. Tree-ensemble architecture's robustness to non-linear relationships and ability to handle structured feature sets allowed them to generalize across subjects more effectively than all other models. The FNN achieved an average accuracy of 72.4% under LOSO conditions, and an average accuracy of 84.5% under pooled 10-fold cross-validation conditions. While it is significantly lower than the accuracy achieved by the top-performing two, it has the advantage of requiring less training time than both other models. It is also worth mentioning that, for the sake of remaining under a certain degree of complexity, the neural networks implements the most basic architecture possible. It doesn't implement any sort of temporal modeling, although that could be taken into account using a Long-Short-term-memory (LSTM) architecture. Implementing such modeling would most likely provide the neural network

with a great advantage on this task, and make it the best performing model in terms of raw accuracy. Lastly, while their performance in terms of accuracy was lower than that of all other models, both KNN and SVM models also came with their unique advantages: KNN achieved very good accuracy (especially on pooled 10-fold c.v.) without requiring any training and only basic tuning, and SVM achieved better accuracy than KNN and higher Macro-F1 scores than the FNN. The classification of the N1 stage remained a big issue across all models. This is likely due to a combination of factors: first of all, as we could see from the EDA it is strongly underrepresented and this, when paired with its EEG pattern that is so similar to that of N2 and wake, probably caused the models to not be able to learn specific characteristics for it. Looking at it from a broad perspective, this project confirms that a carefully engineered feature set combined with lightweight models can provide a good trade-off between performance, interpretability and simplicity, but it also shows the limitations of static, per-epoch models, and definitely keeps the door open for future exploration of sequential architectures, subject-adaptive strategies, better or more complete feature extraction (which we unfortunately didn't have the tools to perform) and better handling of imbalanced data.

This project aimed to identify a low-to-moderate complexity model capable of classifying sleep stages based solely on EEG data and still achieving competitive accuracy (>70%) on generalization tasks. Our approach focused exclusively on the six available EEG channels, from which we extracted 96 features per epoch. These included statistical descriptors and frequency-domain measures. We implemented and evaluated a good variety of models, from simple classifiers like KNN to more powerful ones like XGBoost and FNN, all of which have been tested using cross-validation strategies. The tree-ensemble models emerged as the top-performing models, followed by the FNN, which comes with the disadvantage of being the most complex, SVM, which has a training time that could become prohibitive for some applications, and lastly KNN, which showed the lowest accuracies, while still achieving overall good performance and beating other models on Macro-F1 and training time metrics. The N1 stage remained the most challenging across all models because of its transitional nature and spectral similarity to N2 and wake. However, our results show that it is possible to build efficient and competitive models for sleep stage classification without relying on deep, opaque or overly complex architectures. Although none of our models is perfect, their performance is encouraging and supports the feasibility of using such systems in real-time sleep monitoring contexts. Future improvements could focus on better handling of class imbalance and incorporating lightweight temporal modeling to refine predictions over time, also, extracting features that are more computationally expensive than the ones we extracted might make a strong difference, as we found ourselves to be strongly limited by hardware capabilities.

References

[1] Richard B. Berry, Stuart F. Quan, Alexandre R. Abreu, et al. *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications*. Tech. rep. Darien, IL: American Academy of Sleep Medicine, 2020.

[2] Mary A. Carskadon and William C. Dement. "Normal Human Sleep: An Overview". In: *Principles and Practice of Sleep Medicine*. Ed. by Meir H. Kryger, Thomas Roth, and William C. Dement. 6th. Philadelphia: Elsevier Saunders, 2017, pp. 15–24. DOI: 10.1016/B978-0-323-24288-2.00002-7.

[3] Eduardo López-Larraz et al. *The Bitbrain Open Access Sleep (BOAS) Dataset*. 2024. DOI: 10.18112/openneuro.ds005555.v1.0.0. URL: https://openneuro.org/datasets/ds005555.

[4] Alexander Malafeev et al. “Automatic Human Sleep Stage Scoring Using Deep Neural Networks”. In: *Frontiers in Neuroscience* 12 (2018), p. 781. DOI: 10.3389/fnins.2018.00781.

[5] Huy Phan et al. “SeqSleepNet: End-to-End Hierarchical Recurrent Neural Network for Sequence-to-Sequence Automatic Sleep Staging”. In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 27.3 (2019), pp. 400–410. DOI: 10.1109/TNSRE.2019.2896659.

[6] Rym Nihel Sekkal et al. “Automatic Sleep Stage Classification: From Classical Machine Learning Methods to Deep Learning”. In: *Biomedical Signal Processing and Control* 77 (2022), p. 103751. DOI: 10.1016/j.bspc.2022.103751.

[7] Akara Supratak et al. “DeepSleepNet: A Model for Automatic Sleep Stage Scoring Based on Raw Single-Channel EEG”. In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 25.11 (2017), pp. 1998–2008. DOI: 10.1109/TNSRE.2017.2721116.